

A Platform for Visualizing the Global Environmental Consequences of Nuclear War

Lili Xia, Department of Environmental Sciences, Rutgers University

July 3, 2025

1. Vision:

From abstract threat to local reality: A dynamic platform to visualize nuclear war risk

This project will create an interactive online platform to make the environmental consequences of nuclear war personal, tangible, and undeniable. For a generation that did not live under the constant shadow of the Cold War, the full-scale of nuclear conflict remains a dangerously abstract concept. This is true not only for a full “nuclear winter” scenarios, but also for a regional nuclear war, whose devastating climate consequences would cause worldwide famine and disrupt human society and ecosystems far from the conflict zone. While the public may have a vague sense of immediate destruction, the long-term, cascading global ecosystem collapse remains a dangerously abstract concept. This initiative aims to close that critical knowledge gap. By translating peer-reviewed scientific research into a powerful, story-driven experience, we can catalyze a more urgent and informed global demand for policies that reduce nuclear dangers.

2. The Core Concept:

An Expandable Platform Built on Credible Research

The initiative will develop a user-friendly website that serves as a central hub for visualizing the environmental impacts of nuclear conflict. It will be built as a modular platform, allowing for the progressive addition of new research findings over time.

Module 1: Nuclear Famine

This module will be a direct, interactive translation of Toon et al. (2019), Bardeen et al. (2021), and Xia et al. (2022). The platform is designed as an interactive tool, allowing users to explore the cascading consequences of nuclear conflict. It is structured to answer not only “What would happen?” but, crucially, “Why would it happen?” and “How would it affect me?” The user first starts with a dynamic, high-resolution global map with two initial states:

- Baseline world: A view of global map as it is today, with baseline data on climate and food security
- A world after nuclear war: The user can select from a range of credible conflict scenarios (e.g. a limited regional war, a large-scale exchange) to see the world with nuclear war impact.

Once a scenario is chosen, the user can explore the immediate and long-term environmental shifts. The world map will dynamically visualize the spread of atmospheric soot and the resulting changes in key climate variables. The user can switch between different data layers on the map, such as temperature, precipitation, solar radiation (sunlight), and the surface UV levels. In addition, the user can then select any country (e.g. Japan). The view will shift to a national time-series graph. This will allow them to track the progression of the climate impacts over time at national level.

The core of the experience is to connect the climate variables to its human cost, such as food security. The user can explore the specific impacts on their nation’s food supply after selected nuclear war. They can see visualizations of the projected collapse in different sectors: cereal products, livestock products, and fish products. The platform will present the total available food calories per person in the chosen country, post-conflict. It will then translate this

into a more personal metric: if the nation attempts to feed its population a basic survival-level diet, what percentage of the population would face starvation? This powerful feature makes the abstract concept of “food shortage” deeply and personally resonant.

The platform will also walk the user through the mechanics behind numbers in their chosen country. It begins with their nation’s current reality, visualizing its dietary composition, average caloric intake, and unique food import/export dependencies. It shows a baseline for the population’s needs – the average calories required to maintain normal physical activity based on that country’s specific demographics. With this foundation set, the narrative then directly connects the selected nuclear war scenarios to their plate, illustrating how the modeled climate changes due to nuclear conflict damage their agriculture, pastures and marine fisheries. Finally, the platform will transparently outline the key assumptions in the calculations (e.g. a halt to global trade, levels of domestic food storage, waste reduction) and openly discuss the scientific uncertainties.

3. A platform for growth:

Visualizing other environmental impacts

The platform is designed for growth. After launching the food security module, we will collaborate with leading researchers to develop and integrate additional modules based on established science, such as UV impact on human health and global water contamination.

References:

- Toon, Owen B., Charles G. Bardeen, Alan Robock, Lili Xia, Hans Kristensen, Matthew McKinzie, R. J. Peterson, Cheryl Harrison, Nicole S. Lovenduski, and Richard P. Turco, 2019: Rapid expansion of nuclear arsenals by Pakistan and India portends regional and global catastrophe. *Science Advances*, **5**, eaay5478, doi:10.1126/sciadv.aay5478.
- Bardeen, Charles G., Douglas E. Kinnison, Owen B. Toon, Michael J. Mills, Francis Vitt, Lili Xia, Jonas Jägermeyr, Nicole S. Lovenduski, Kim J. N. Scherrer, Margot Clyne, and Alan Robock, 2021: Extreme ozone loss following nuclear war resulting in enhanced surface ultraviolet radiation. *J. Geophys. Res. Atmos.*, **126**, e2021JD035079, doi:10.1029/2021JD035079.
- Xia, Lili, Alan Robock, Kim Scherrer, Cheryl S. Harrison, Benjamin Leon Bodirsky, Isabelle Weindl, Jonas Jägermeyr, Charles G. Bardeen, Owen B. Toon, and Ryan Heneghan, 2022: Global food insecurity and famine from reduced crop, marine fishery and livestock production due to climate disruption from nuclear war soot injection. *Nature Food*, **3**, 586-596, doi:10.1038/s43016-022-00573-0.